

Less-acidic boric acid-functionalized self-assembled monolayer for mitigating NiO_x corrosion for efficient all-perovskite tandem solar cells

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The interfacial contact between NiO_x and self-assembled monolayers (SAMs) in wide-bandgap (WBG) subcells limits the efficiency and stability of all-perovskite tandem solar cells (TSCs). The strongly acidic phosphoric acid (PA) anchors in common PA-SAMs corrode reactive NiO_x , undermining device stability. Moreover, SAM aggregation leads to interfacial losses and significant open-circuit voltage (V_{OC}) deficits. Here, we introduce boric acid (BA) as a milder anchoring group that chemisorbs onto NiO_x via strong $-\text{BO}_2^-$ –Ni coordination. A benzothiophene-fused head group enhances interfacial bonding through S–Ni orbital interactions, yielding higher binding energy than PA-SAMs. This design also promotes homogeneous SAM formation without aggregation. Resultantly, the WBG cell exhibits an improved PCE to 20.1%. When integrated with narrow bandgap (NBG) subcell, the two-terminal (2T) TSCs achieve an ameliorative PCE of 28.5% and maintain 90% of the initial PCE after maximum power point tracking (MPP) under 1 sun illumination for 500 h.

All-perovskite tandem solar cells (TSCs), consisting of a wide-bandgap (WBG, 1.7–1.8 eV) top cell paring with a narrow bandgap (NBG, 1.2–1.3 eV) bottom cell, have been presented as a promising approach to breaking the Shockley-Queisser (SQ) limits of single-junction perovskite solar cells (PSCs)^{1–3}. With the rapid advancements of subcells and interconnection layers, TSCs have reached certified power conversion efficiency (PCE) of 30.1%, demonstrating the great potential to be commercialized as cost-effective photovoltaic (PV) technology^{4,5}.

Whereas, photovoltaic efficiency and stability of all-perovskite TSCs are both severely limited by the suboptimal interfacial contacts

between NiO_x and self-assembled monolayers (SAMs) in WBG subcell. Though SAMs directly coating on transparent conductive oxides (TCO, including ITO or FTO) substrates demonstrates a universal way for the development of single-junction inverted PSCs^{6,7}, NiO_x /SAM combination has been generally utilized as a hole-transporting layer (HTL) in WBG cells and TSCs^{8–11}. This might be due to the commonly observed wrinkle morphological features of the buried WBG perovskite with much higher roughness than that of 1.55 eV bandgap perovskite (low content of Cs and Br). Therefore, NiO_x is required to prevent current leakage between perovskite/ITO or FTO electrodes, as a single SAM

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layer is too thin to avoid shunting. While, due to the higher reactivity of NiO_x than ITO and FTO^{12,13}, the strong acidic phosphoric acid (PA) anchor in the widely employed PA-functionalized SAMs would corrode NiO_x , which is detrimental to the long-term stability of solar cell devices. Additionally, the easy agglomeration of the commonly used PA-functionalized SAM leads to unsatisfied surface coverage and some weakly anchored SAMs through $\text{OH}\cdots\text{O}=\text{P}$ hydrogen bonding. Such loosely bonded sites are prone to be desorbed by strong polar solvents such as DMF¹⁴, resulting in a random redeposition at the perovskite bottom layer. It not only leads to nonradiative recombination and substantial V_{OC} deficit^{15–18}, but also severely limits the operational stability of PSCs.

Besides, WBG perovskite with a high Br content (40%) suffers from an inhomogeneous nucleation-crystallization process, characterized by a markedly rapid crystallization rate for Br-rich perovskite. This phenomenon may stem from the different coordination strength between FA-DMSO and $\text{PbI}_2/\text{PbBr}_2$ -DMSO adducts^{19–21}. Consequently, nonuniform I-rich and Br-rich regions are formed, which further lead to trap-assisted nonradiative recombination centers. Such inferior film morphologies are also susceptible to phase segregation under continuous illumination, compromising the operational stability of WBG cells^{22,23}.

Previous studies have shown that SAMs could play multiple roles in the performance of PSCs, that is strengthening the bridging bonds between SAM and metal oxide substrates (ITO, FTO, NiO_x), and simultaneously regulating the crystallization of WBG perovskite films^{24–27}. A commonly employed strategy to address issues related to SAM wettability and agglomeration in inverted PSCs with a bandgap of 1.55 eV is the co-assembly of SAM with small molecules on ITO or FTO. This approach improves surface coverage and morphology of the buried perovskite layer^{10,14,15,25–28}. It is important to note that while PA-based SAMs have been widely used to modify the surface of ITO or FTO^{29,30}, they do not necessarily fit on NiO_x substrate, considering the higher reactivity of NiO_x than ITO/FTO. This can cause corrosion at the interface, leading to stability issues¹³, particularly when upscaling device fabrication, where TCO substrates necessitate rinsing in SAM solution for the anchoring process³.

Herein, in this work, the acidic-weakened boric acid functionalized SAM (BA-SAM) with various fused cores are reported to anchor onto the NiO_x surface. Density functional theory (DFT) calculations and X-ray photoelectron spectroscopy (XPS) analysis reveal strong coordination between $-\text{BO}_2^-$ and Ni. The design of benzothiophene fused core as the functional head group strengthened the interfacial bonding through an additional orbital-pair contribution from S-Ni interaction, resulting in a higher binding energy (−6.73 eV) than that of PA-SAM (−6.14 eV) on NiO_x . Such interaction between the fused head core and NiO_x surface further benefits the homogeneous formation of BA-functionalized SAM on the NiO_x surface without aggregation. These expect to improve the surface coverage and enhance the interfacial stability of the WBG subcell. Additionally, we elucidate a π -cation interaction between benzothiophene fused core and FA^+ cation through a combination of theoretical and experimental analysis, which engenders a balanced crystallization rate of I-rich and Br-rich perovskite phases. The resulting WBG perovskite film presents a homogeneous I/Br distribution and mitigated phase segregation under continuous illumination aging. By further mixing BA-SAM with a small amount of the commonly utilized Me-4PACz (at a molar ratio of 4:1, BA-SAM: Me-4PACz), the WBG cell shows a substantial improvement in PCE from 18.9% (control device based on Me-4PACz) to 20.1%, with V_{OC} of 1.30 V, J_{SC} of 18.2 mA cm^{-2} and FF of 84.8%. Integrating with the NBG subcell, the 2T TSCs achieve an ameliorative PCE of 28.5%, along with notable operational stability by retaining 90% of the initial PCE after maximum power point tracking (MPP) under 1 sun illumination for 500 h.

Results

Adsorption mechanisms of SAMs on NiO_x surface

To dig into the adsorption mechanism of different SAMs on NiO_x surface, DFT calculations including the electron localization function (ELF) were performed on Me-4PACz/ NiO_x , Me-4BACz (substituting PA with BA)/ NiO_x , S-BA/ NiO_x and O-BA/ NiO_x , N-BA/ NiO_x models (Supplementary Fig. 1 and Fig. 1a–d). Noted that S-BA, O-BA, and N-BA are BA-functionalized SAMs with benzothiophene, pyridine, and benzofuran cores, respectively. Firstly, we analyze the adsorption mechanism of BA on NiO_x by comparing Me-4PACz/ NiO_x and Me-4BACz/ NiO_x models, as shown in Supplementary Fig. 1f, g. Figure 1b demonstrates an obvious symmetry breaking and deformation between $-\text{BO}_2^-$ and Ni, suggestive of orbital overlapping of O-Ni bonding. The differential charge density mapping in Supplementary Fig. 1m exhibits charge transfer between $-\text{BO}_2^-$ and Ni as well. This is further confirmed by the Crystal Occupied Hamilton Population (COHP) calculation. Illustrated by Fig. 1e, the interactions between Ni 3d-O 2p contribute to the formation of the bonding state, while the anti-bonding state originates from Ni 4s-O 2s2p and Ni 3d-O 2p. The efficient anchoring could be attributed to the high Lewis acidity of NiO_x , which facilitates a dominate coordination process. In contrast, on less Lewis acidic ITO substrates, the adsorption mechanism of SAM involves heterocondensation first, primarily affected by pKa of the tailoring group⁶. Such orbital-pair contribution compensates for the relatively weak acidity of the BA group, generating a comparable binding energy of Me-4BACz (−5.58 eV) than that of Me-4PACz (−6.14 eV) (Supplementary Table 1). The results are consistent with the previous report on BA-based SAM³¹.

To further study the effects of fused rings on the adsorption, ELF images of S-BA/ NiO_x were plotted. From Fig. 1b, a symmetry breaking and deformation between S (from S-BA) and Ni (from NiO_x) is unveiled, illustrating a charge transfer between the S atom and NiO_x . Similarly, from Fig. 1f, the COHP shows a bonding state from Ni 4s-S 3p and an anti-bonding state from Ni 3d4s-S 3p. This interaction, which is absent in Me-4PACz/ NiO_x , further contributes to strengthening the bonding between the S-BA and NiO_x surface. As a result, the S-BA shows stronger anchoring than Me-4PACz on the NiO_x surface, with a more negative binding energy of −6.73 eV. Comparatively, O-BA and N-BA show less negative binding energies of −6.67 and −6.25 eV, respectively. These results confirm the robust anchoring of S-BA on NiO_x .

On top of that, as oxygen vacancies are also active sites for SAM adsorption¹⁴, the adsorption models of different SAMs on oxygen-deficient NiO_x surfaces were also calculated. From the results in Supplementary Fig. 2, the binding energies can be summarized in Supplementary Table 2. It is seen that S-BA also shows a more negative binding energy of −6.53 eV than the Me-4PACz analog (−6.34 eV), confirming the preferred adsorption of S-BA on oxygen-deficient NiO_x .

Hole conducting characteristics of the above BA-tailed SAMs were then investigated by quantum chemical calculations. From the frontier molecular orbitals in Supplementary Fig. 3, it is observed that the electron-rich carbazole dominates both HOMO and LUMO levels in Me-4PACz and Me-4BACz, delivering high-lying LUMO levels of −0.92 eV and −0.76 eV, respectively. The high excited state energy of carbazole under illumination makes it susceptible to chemical reactions. It would lead to light-induced degradation, which is disadvantageous to interfacial contacts and charge transfer between SAM and NiO_x ³². Contrastingly, Supplementary Fig. 3 shows a high level of delocalization of HOMO and LUMO levels near the S atom and anchoring group. Such delocalization is even more intense in the mixed S-BA and Me-4PACz system (denoted as S-BA-SAM), as shown in Supplementary Fig. 3, which is favorable for electron/charge delocalization and transport, as well as a more stable LUMO level (S-BA-SAM: −1.45 eV vs. Me-4PACz −0.92 eV). The results are consistent with the experimental findings that mixing S-BA with a small content of

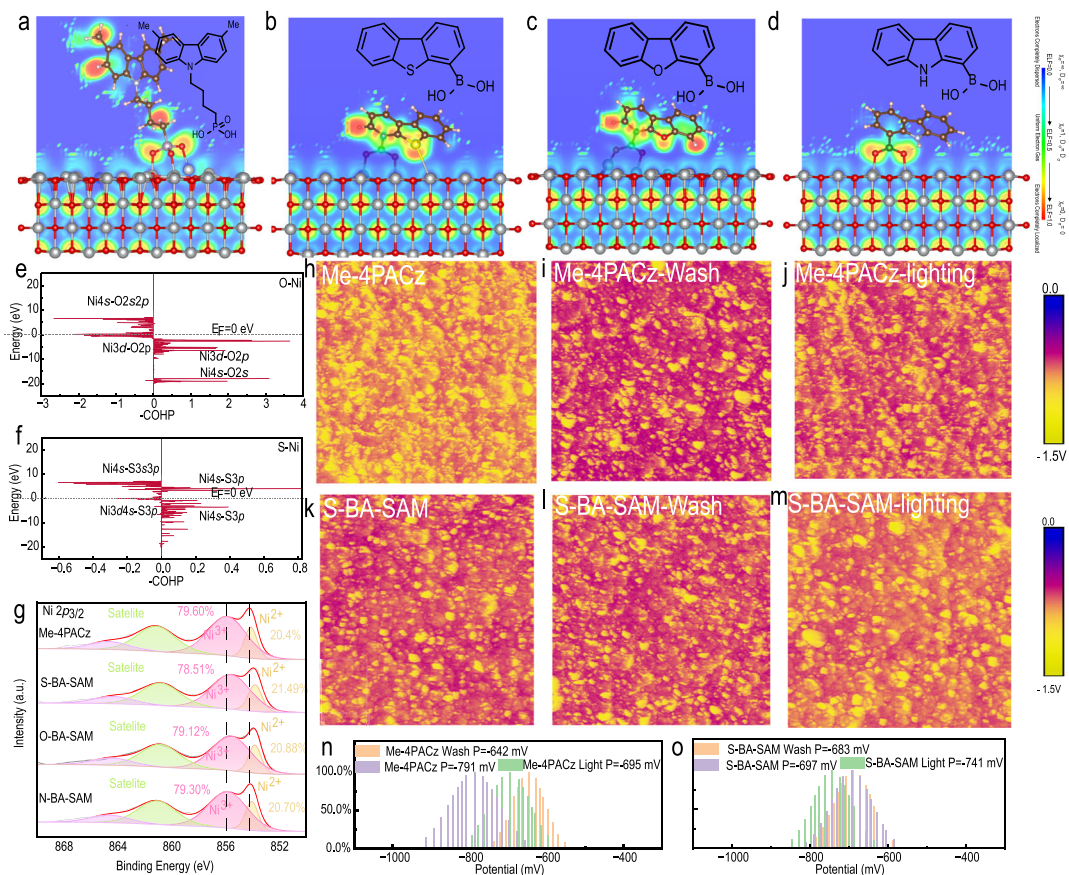


Fig. 1 | Anchoring of different SAMs on NiO_x . ELF images absorbed by (a) Me-4PACz, (b) S-BA, (c) O-BA, (d) N-BA on the NiO_x surface were obtained by DFT calculation. COHP analysis of (e) O-Ni and (f) S-Ni bonding between S-BA and NiO_x surfaces. (g) The Ni 2p_{3/2} XPS spectra of the NiO_x film with SAMs (Me-4PACz, S-

BA-SAM, O-BA-SAM and N-BA-SAM). KPFM images of (h) the NiO_x /Me-4PACz original film and (i) the film after DMF washing and (j) the film after 6 h of light; (k) the NiO_x /S-BA-SAM original film and (l) the film after DMF washing and (m) the film after 6 h of light. CPD distributions of (n) NiO_x /Me-4PACz and (o) NiO_x /S-BA-SAM.

Me-4PACz (at a molar ratio of 4:1, BA-SAM: Me-4PACz) benefits the hole-transporting process, in terms of achieving larger conductivity, higher hole mobility, and improved photovoltaic efficiency, which will be discussed in the later context. Therefore, in the following experimental studies, the mixed BA-functionalized SAM with Me-4PACz were characterized in comparison with Me-4PACz.

Noted that DFT calculations were again adopted to study the bonding configurations of mixed SAM, in an attempt to unveil the anchoring competition process between BA- and PA-tailed molecules in the mixed SAM system. ELF diagram of S-BA-SAM in Supplementary Fig. 4 shows the deformation of Ni upon its interaction with O and S, evidencing the chemical interactions of Ni-O and Ni-S. This supports that S-BA remains binding to NiO_x through the original three-site anchors (two covalent B-O bonds and one S-Ni interaction), even under the coexistence of Me-4PACz. Similar results are also observed in the oxygen-deficient NiO_x system (Supplementary Fig. 5). Therefore, it is proved that Me-4PACz would not cause the desorption of BA-based SAM.

The anchoring strength of SAMs on NiO_x was then evaluated using XPS measurements, with key parameters summarized in Supplementary Table 3. Firstly, it is noted that the construction of monolayer SAM in PSC devices is actually an ideal case. Han et al. pointed out that highly efficient PSCs require a deposited SAM thicker (~6 nm) than one monolayer, because the deposition of perovskite layer (DMF solvent) would wash away some SAM molecules. On top of this, most studies in high-impact journals did not incorporate a post-cleaning treatment in their experimental procedures^{3,8,27,33–35}. Even when post-treatment was

mentioned^{36,37}, it involved spin-casting ethanol onto the SAM film, instead of rinse. As the former would just wash away some molecules on the top layer, while the latter leads to the formation of exact “monolayer”, which is not good for device performance. Following this spin-coating cleaning treatment, the NiO_x /SAM films were post-cleaned with ethanol through spin-casting before test. As shown in Fig. 1g, the XPS spectra of NiO_x /Me-4PACz film exhibit Ni 2p_{3/2} peaks at 855.9 and 854.1 eV, corresponding to Ni^{3+} and Ni^{2+} , respectively. With the co-SAM strategy by S-BA-SAM and O-BA-SAM, Ni 2p_{3/2} spectra shift downward by 0.4 eV and 0.2 eV, respectively, while that of NiO_x /N-BA-SAM shows negligible shifts. The significant downward shifts in NiO_x /S-BA-SAM are illuminative of facilitated charge transfer between NiO_x and S-BA-SAM, suggesting an increased anchoring strength of S-BA-SAM on NiO_x ^{38,39}. This enhanced interfacial bonding is further supported by a reduction in the Ni^{3+} ratio, from 79.6% in NiO_x /Me-4PACz film to 78.5% with S-BA-SAM and 79.1% with O-BA-SAM. It reflects an improved surface chemical environment, which benefits the long-term stability of the buried interface³⁹.

The corrosion effects of the SAM on NiO_x were then studied, which could result from two processes. First, the solution deposition process of SAM onto NiO_x exposes the NiO_x to an acidic environment, leading to a corrosive reaction of $\text{NiO}_x + 2\text{H}^+ \rightarrow n\text{Ni}^{2+} + (1-n)\text{Ni}^{3+} + \text{H}_2\text{O}$. To quantify the leaching of Ni^{2+} and Ni^{3+} ions, ITO/ NiO_x substrates were immersed in S-BA-SAM or Me-4PACz solution for 5 h, followed by detection of the Ni ion concentration using Inductively Coupled Plasma Optical Emission Spectrometer (ICP-OES). As shown in Supplementary Fig. 6, the concentration of Ni ions in S-BA-SAM solution (29 $\mu\text{g/L}$) is

significantly lower than that in Me-4PACz solution (87 $\mu\text{g/L}$), confirming the suppressed corrosion of NiO_x by less acidic BA-functionalized SAM. Second, it is unveiled from Fourier transform infrared (FTIR) spectroscopy (Supplementary Fig. 7) that considerable amount of PA-tails fail to bind to NiO_x through P-O-Ni covalent bonds (with strong P-OH signal existed), suggesting a substantial presence of phosphoric acid groups on the NiO_x surface, aligning with other published reports⁴⁰. XPS analysis further quantifies the number of covalent bonds in S-BA-SAM/ NiO_x and Me-4PACz/ NiO_x films. As illustrated in Supplementary Fig. 8a, b, Me-4PACz/ NiO_x exhibits a low P-O-Ni to P-O-H ratio of only 27%, supporting the presence of significant acid groups on the surface. Contrastingly, the S-BA-SAM/ NiO_x shows notably higher B-O-Ni to B-O-H ratio of 47%. This might be attributed to the easier aggregation of Me-4PACz, which will be elaborated upon later. The presence of -POOH on the surface can corrode the NiO_x substrate, particularly under thermal or light accelerated aging, which is again studied by ICP-OES. To obtain measurable results, 8 pieces of $2\text{ cm} \times 2\text{ cm}$ Me-4PACz/ NiO_x substrates were aged under light illumination for 40 h, followed by immersion in ethanol solvent for 20 min to wash out the leached Ni ions. As shown in Supplementary Fig. 9, a considerable amount of Ni ions (32 $\mu\text{g/L}$) was detected, verifying the corrosive reaction. In contrast, the S-BA-SAM/ NiO_x substrates aged under the same condition show negligible content of Ni ions, suggesting enhanced interfacial stability.

The corrosive effects of -POOH on NiO_x in thin film states were further analyzed by XPS spectra, where the peak area ratio of hydroxide and lattice O atoms (Ni-O) in O 1s core was determined¹⁴. As shown in Supplementary Fig. 10, the ratio of hydroxide and lattice O atoms in the NiO_x /Me-4PACz film after light aging increases significantly from 82% to 122%, suggesting the increment of unbonded -OH and decrement in covalent Ni-O on NiO_x surface. The results indicate deteriorated interfacial contact with significant desorption of Me-4PACz from the NiO_x surface, which could be due to corrosive effects of unanchored -POOH on NiO_x in thin film states. Aging tests on ITO/Me-4PACz samples were also conducted to have a comparison with those on NiO_x substrate. From XPS spectra shown Supplementary Fig. 11, ITO/Me-4PACz shows slightly increased peak area ratio of hydroxide and lattice O atoms from 84% to 102% after 5 h light aging, further suggesting the higher reactivity of NiO_x than ITO and the urgency of using weak-acid anchors. Using the BA-based SAM, the ratio in the NiO_x /S-BA-SAM shows marginal changes (from 68% to 63%) under light aging, suggesting the less-acidic BA-SAM helps maintain the integrity of the interfacial contact.

The influence of post-cleaning step and residual SAM molecules on the corrosion process were then analyzed by XPS measurements on NiO_x /SAMs before and after ethanol washing. The XPS analysis (Supplementary Figs. 12, 13 and Supplementary Table 4) reveals a higher surface coverage for the unwashed sample (1.22×10^{-2}) compared to the washed ones (1.02×10^{-2}), indicating that a portion of residual molecules was removed from the top surface. Noted that even with the cleaning step (ethanol spin-casting rather than rinsing), a film thicker than a monolayer is formed, which is recognized to be critical for the fabrication of highly efficient PSCs^{3,8,14,27,33,34}. After 5-h light aging, the unwashed ITO/ NiO_x /Me-4PACz exhibits a comparable increment (from 96% to 145%) in the peak area ratio of hydroxide to lattice O atoms relative to the washed ones (from 82% to 122%), as shown in Supplementary Fig. 14. This suggests that the additional unbonded -POH groups on the top surface have a limited impact on the interfacial stability, implying that the molecules near the buried interface play a more significant role in the corrosion process. A similar trend was also observed in ITO/ NiO_x /S-BA-SAM films as well (Supplementary Fig. 15). This finding aligns with our observation of similar device stabilities for PSCs with or without washing step.

Another critical issue raised by the traditional Me-4PACz is its tendency to aggregate or crystallize during solution deposition, driven

by strong van der Waals interactions, particularly π - π interactions. This has also been testified by molecular dynamic (MD) calculations in Supplementary Fig. 16, which shows the formation of Me-4PACz dimers, trimers, and tetramers aggregations. The molecular aggregation would lead to unsatisfied surface coverage and some weakly anchored SAMs through $\text{OH} \cdots \text{O}=\text{P}$ hydrogen bonding. Such loosely bonded sites could be desorbed by strong polar solvents such as DMF¹⁴, resulting in random redeposition at the perovskite bottom layer. It expects to severely limit the operational stability of PSCs. S-BA-SAM, on the other side, shows homogeneous distribution without aggregation, which might be due to the strong interaction between the benzothiophene on S-BA and NiO_x , as well as the less π - π interactions through benzothiophene, as illustrated by the MD calculations in Supplementary Fig. 16b.

Correspondingly, the resistance of two acid-functionalized SAMs to DMF was thus tested. The NiO_x /SAM samples rinsed in DMF solvent for different periods were prepared, which were subsequently measured by XPS. Samples are not exposed to ambient air at any time. Robustness of SAM (S-BA-SAM and Me-4PACz) on the NiO_x surface upon DMF washing could be determined by the coverage factor, which was calculated as the core level area of C1s in SAMs molecule normalized to the NiO_x 3p3/2 core level area in XPS⁴¹. From the C 1s and Ni 2p3/2 XPS spectra in Supplementary Figs. 17–22, surface coverage could be summarized in Supplementary Fig. 23 and Supplementary Tables 5, 6. S-BA-SAM shows significantly higher coverage factors under a variety of DMF washing volumes (0–1200 μL) than those of Me-4PACz, evidencing a decreased desorption of S-BA-SAM from NiO_x during the deposition process of perovskite film.

The less corrosive effect, together with improved robustness of S-BA-SAM on NiO_x , is expected to improve the interfacial contact and its long-term stability. Subsequently, the structural and electronic properties of NiO_x /SAM were assessed before and after light aging and DMF washing. As shown in Supplementary Fig. 24, the Raman peak at 793 cm^{-1} belongs to the B-O bond⁴². Other peaks near 1000–1400 cm^{-1} (C-H and C-C stretching vibrations in the methyl segment) and 1600 cm^{-1} (C-C stretching in benzothiophene) confirm the presence of benzothiophene^{43,44}. As shown in Supplementary Fig. 24a–c, the characteristic Raman peaks of the sample show minor changes after DMF washing and illumination for 12 h. While in the NiO_x /Me-4PACz system, the intensity is greatly reduced, indicating that BA-functionalized SAM possesses higher structural stability.

Similarly, the surface potential changes of NiO_x /SAM layers were inspected by Kelvin probe force microscopy (KPFM). As shown in Fig. 1h–o, the contact potential difference (CPD) distribution for S-BA-SAM is narrower than that of the control sample, indicative of the homogeneous formation of S-BA-SAM on NiO_x substrate. Additionally, mixed SAM strategy shifts the CPD peak from -791 mV (control) to less negative values of -697 mV, -741 mV, and -742 mV for S-BA-SAM, O-BA-SAM and N-BA-SAM, respectively. This suggests a reduction in the work function of the NiO_x /SAM layer, as described by the equation $\text{CPD} = (\Phi_{\text{tip}} - \Phi_{\text{sample}})/e$ ⁴⁵. This effect might be ascribed to the less electronegative BA-tail for lowering the p-type characteristics of the surface. The results are consistent with the energy level alignment extracted from ultraviolet photo-electron spectroscopy (UPS) measurements, as illustrated in Supplementary Figs. 25–28. Moreover, the control film (NiO_x /Me-4PACz) shows a substantial CPD shift of 167 mV (from -791 mV to -642 mV) upon DMF washing, illustrating a weak adsorption of Me-4PACz on NiO_x due to molecular aggregation. Conversely, NiO_x /S-BA-SAM shows negligible CPD shift before and after washing, concealing stronger bonding interactions. As a result, light aging (continuous illumination for 6 h) imposes a minimal impact on the surface potential variations of the NiO_x /S-BA-SAM layer. In addition to that, conductivities of NiO_x /SAMs were further measured. From Supplementary Fig. 29, NiO_x /S-BA-SAM film shows larger conductivity ($3.84 \times 10^{-3} \text{ S m}^{-1}$) than that of NiO_x /Me-4PACz ($3.16 \times 10^{-3} \text{ S m}^{-1}$),

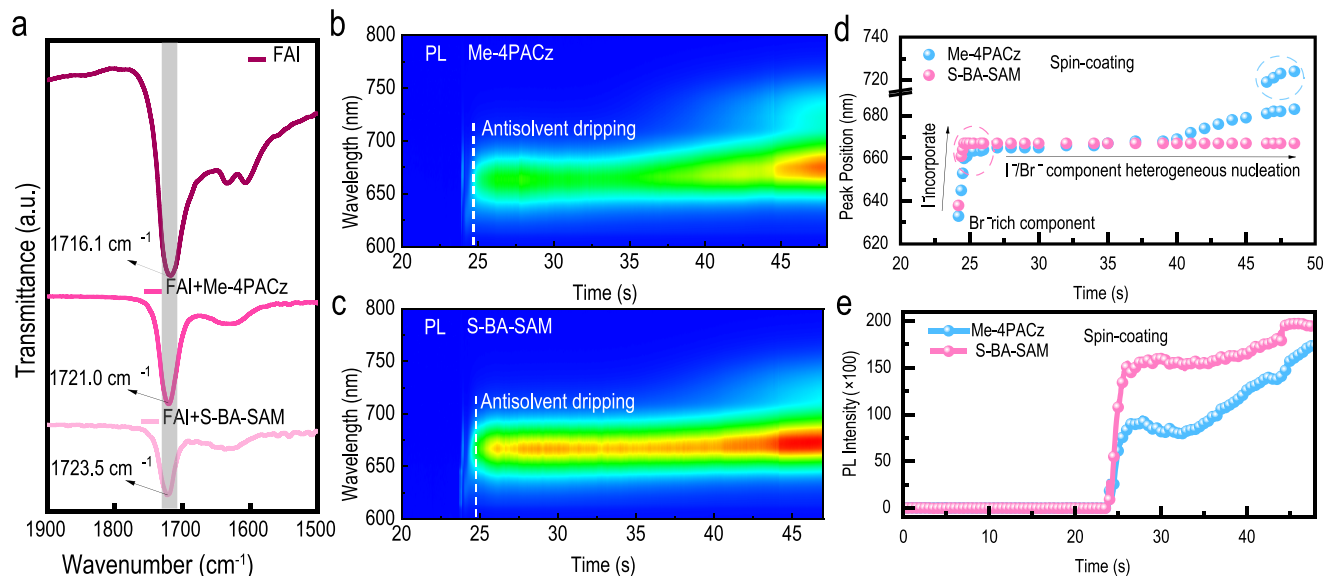


Fig. 2 | Effects of different SAM on regulation the nucleation of perovskites. (a) FTIR of FAI, FAI + Me-4PACz and FAI + S-BA-SAM. In situ PL spectroscopy of perovskite film during spin-coating with (b) Me-4PACz and (c) S-BA-SAM modification.

(d) Peak position variations of Me-4PACz and S-BA-SAM modified perovskite during spin coating. (e) PL intensity variations of Me-4PACz and S-BA-SAM modified perovskite during spin coating.

indicative of improved hole conducting ability. Its conductivity remains almost constant before and after illumination, while NiO_x/Me-4PACz shows a significant reduction to $1.91 \times 10^{-3} \text{ S m}^{-1}$ after 12 h of illumination. The above results demonstrate enhanced structural and electronic stability of S-BA-SAM.

Effects of SAMs on perovskite crystallization

FTIR spectra of FAI, FAI + SAM, and SAM were recorded to dig into the π -cation interactions between FA⁺ cation and SAM. From Fig. 2a and Supplementary Fig. 30, it is seen that the skeletal vibration of FA⁺⁴⁶, shifts from 1716.1 cm⁻¹ to 1721 cm⁻¹, 1723.5 cm⁻¹, 1716.1 cm⁻¹ and 1718.0 cm⁻¹, respectively, for Me-4PACz, S-BA-SAM, O-BA-SAM and N-BA-SAM. The shifts of FTIR peaks to higher wavenumbers could be explained by the electron-withdrawing effects of the fused rings. Benzothiophene fused ring, with the strongest electron capture ability, imparts the large force constant to the FA⁺ cation (benzothiophene > pyridine > benzofuran)⁴⁷, resulting in the highest vibrational frequency. Therefore, the energy required for the transition from the ground state to the first excited state (i.e., the highest vibrational frequency) is greatest for the FAI/S-BA-SAM sample. The results testify the strong π -cation interactions between FA⁺ and S-BA-SAM.

Such interactions tend to slow the crystallization rate of both FAI/PbI₂ and FAI/PbBr₂, thereby inducing a homogeneous formation of I-rich and Br-rich regions. To verify this, in-situ UV-Vis absorption spectroscopy and photoluminescence (PL) measurements were performed, enabling real-time monitoring of perovskite nucleation and crystallization during spin-coating and thermal annealing, as depicted in Fig. 2b, c and Supplementary Fig. 31. From the 2D pseudo-color plots (Fig. 2b, c), the PL peak positions and intensities over time provide critical insights. Clear PL signals are detected immediately upon antisolvent dripping at ~24 s, indicative of the onset of perovskite nucleation. As shown in Fig. 2d, PL signals in both films initially appear at short wavelengths (630–640 nm) and then redshift to ~660–670 nm over 24–26 s. This reflects that the nucleation of the Br-rich region occurs first in the WBG perovskite film, followed by I⁻ diffusion into the Br-rich nuclei, realizing the mixed I/Br nucleation process⁴⁸. Notably, S-BA-SAM treatment accelerates the nucleation of the I-rich

component, as highlighted by the red-dotted circle at ~25 s. It delivers heterogeneous nucleation of mixed I/Br halide perovskite, stabilizing the WBG perovskite phase during spin-coating (~25–50 s). Contrastingly, the control WBG film deposited on Me-4PACz presents an uncontrollable nucleation manner with spontaneous phase segregation of the I-rich region and Br-rich region. This engenders non-radiative recombination and poor light stability of WBG perovskite film, negatively affecting both photovoltaic efficiency and operational stability of the subcell devices⁴⁸. Additionally, it can be seen from Fig. 2e that the PL intensity of S-BA-SAM is higher than that of Me-4PACz. In situ UV-Vis absorption spectra during spin-coating (Supplementary Fig. 31a, b) align well with the in situ PL spectra (Supplementary Fig. 31c, d). Specifically, S-BA-SAM treated perovskite film shows a rapid absorption band-edge expansion upon antisolvent dripping, suggesting a fast and homogenous nucleation of I/Br mixed halides. The rapid nucleation further gives rise to densely formed perovskite nuclei in the target film, which could be reflected by the significantly higher absorbance in S-BA-SAM treated perovskite film than that of the control ones. The PL spectroscopy of both Me-4PACz and S-BA-SAM treated perovskite films during thermal annealing are shown in Supplementary Fig. 32. It is seen that the S-BA-SAM treated sample shows a gradual PL intensity increment during the first 4 s, while that of the control sample instantly reaches the highest value. It suggests a slower crystal growth of perovskite, which assists in the formation of higher-quality perovskite film⁴⁹. The attenuated PL intensity in both samples could be explained by the unavoidable film degradation under intense incident light excitation.

Since the crystallization process typically begins at the air/liquid surface, a faster Br-rich region crystallization rate over the I-rich one leads to a nonuniform vertical distribution of Br and I, with a large amount of Br frozen near the film surface^{50–52}. This is evidenced by the time-of-flight secondary ion mass spectrometry (TOF-SIMS) analysis of the whole devices. As shown in Supplementary Fig. 33a, b, the mixed SAM strategy results in a lower Br/I ratio in the target film than that in the control film, further illustrating a homogeneous distribution of halide components.

To study whether the crystallization changes were caused by the wettability of the buried layer, contact angles of perovskite precursors

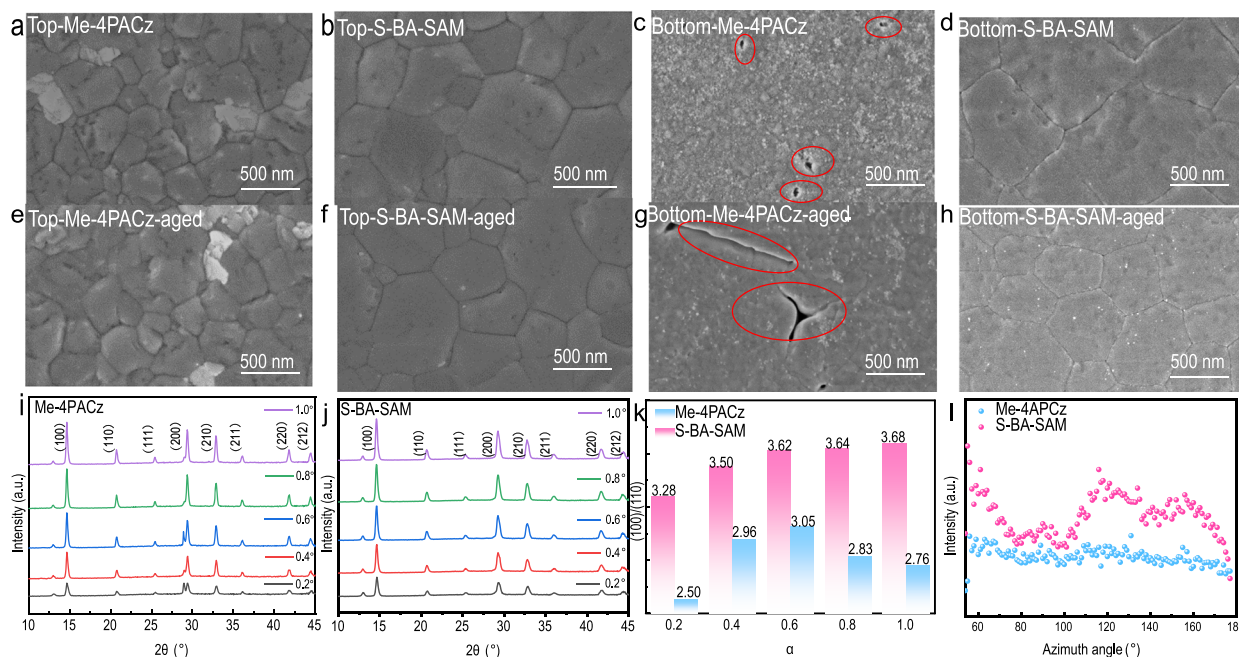


Fig. 3 | Effects of different SAMs on the morphology of perovskite films. SEM images of the top surface of perovskite films on (a) Me-4PACz and (b) S-BA-SAM; SEM images of the buried interface morphology of perovskite films on (c) Me-4PACz and (d) S-BA-SAM; SEM images of the top surface of perovskite films on (e) Me-4PACz and (f) S-BA-SAM after 6-h light aging; SEM images of the buried interface morphology of perovskite films on (g) Me-4PACz and (h) S-BA-SAM after 6-h light

aging. GIWAXS of perovskite films with (i) Me-4PACz and (j) S-BA-SAM modification with a variety of incident angles (α) 0.2°–1.0°. (k) Ratio of (100) to (110) diffraction peak intensities of perovskite films with Me-4PACz and S-BA-SAM modification with a variety of incident angles (α) 0.2°–1.0°. (l) Integrated azimuth angle at 14.24° (100) from GIWAXS pattern in Supplementary Figs. 37 and 38 ($\alpha = 0.2^\circ$).

on NiO_x/SAM were measured. Presented by Supplementary Fig. 34, the Me-4PACz film exhibits a mean contact angle of 86.3°, whereas the S-BA and S-BA-SAM film demonstrate slightly lower but comparable contact angles of 68.3°, and 72.3°, respectively. The slight decrement might be due to the homogeneous distribution of SAM without aggregation (as evidenced by MD calculation and KPFM results). While the core group of S-BA (benzothiophene) is still hydrophobic, both pure S-BA and its mixture with Me-4PACz (S-BA-SAM) do not show a significant reduction in contact angle values. As a result, we conclude that the primary influence of buried SAM on perovskite crystallization arises from the π -cation interactions between benzothiophene and FA⁺.

The chemical environment of the WBG perovskite surfaces was subsequently studied by XPS measurements. Illustrated in Supplementary Fig. 35a, Pb 4f_{5/2} and Pb 4f_{7/2} peaks shift downward from 143.2 and 138.4 eV (Me-4PACz) to 143.0 and 138.2 eV for S-BA-SAM, suggestive of a reduction in undercoordinated Pb²⁺. While those peaks show marginal shifts for O-BA-SAM and upshift to 143.4 and 138.6 eV for N-BA-SAM. This might be attributed to the more balanced crystallization of Br/I halide in S-BA-SAM treated film, which effectively suppresses the phase segregation, particularly at the film surface (as evidenced by the TOF-SIMS result). As phase segregation typically induces tensile strains in perovskite lattices or grains, which degrades the film surface and exposes undercoordinated Pb²⁺. Improved crystallization, as seen in the S-BA-SAM treated perovskite film, could stabilize the perovskite lattice and strengthen the N-H...I hydrogen bonds. This is evidenced by the downshift in the I 3d XPS spectra of S-BA-SAM treated perovskite film, as illustrated in Supplementary Fig. 35b.

Effects of the modulated crystallization kinetics on the top and bottom surface morphologies were directly assessed by scanning electron microscopy (SEM) images. Among the top-surface morphologies of WBG perovskite on different SAMs illustrated in Fig. 3a, b and

Supplementary Fig. 36, the S-BA-SAM-based target film demonstrates the largest and most compact grains without unreacted PbI₂ regions or pinholes. Similarly, the buried interface of S-BA-SAM/perovskite film shows greatly improved morphology compared with the control film, with markedly enlarged grain size and the elimination of numerous nanovoids (Fig. 3c, d). Such features render improved resilience of perovskite films to light illumination. Contrary to the severe degradation of the control film, the target films show negligible morphological changes (Fig. 3e–h).

To scrutinize the crystallographic orientation from the surface to the bulk film along the vertical direction, grazing incident wide-angle X-ray scattering (GIWAXS) with a variety of incident angles 0.2°–1.0° were illustrated in Supplementary Figs. 37, 38 and Fig. 3i–k. From Fig. 3i, the control film on top of Me-4PACz exhibits a noticeable splitting of the diffraction peaks from the (200) lattice plane (−28.9°), which might be ascribed to the lattice distortion caused by residual stresses. Such a phenomenon disappears in the target film based on S-BA-SAM (Fig. 3j). Additionally, the target film shows an overall increased intensity ratio of the (100) and (110) diffraction peaks ($I_{(100)}/I_{(110)}$), as illustrated in Fig. 3k. It suggests that the mixed SAM strategy assists a preferred orientation of the (100) lattice plane throughout the entire film, which would benefit charge carrier transport across the perovskite lattices. Whereas, $I_{(100)}/I_{(110)}$ values are significantly lower on both the top surface and the bottom layer in the control film. The reduced top-surface value might be due to uncontrolled Br/I halide crystallization kinetics, while the low value at the bottom layer might result from poor surface coverage of SAM and nanovoids formed at the buried interface. Additionally, azimuthal integration analysis of the (100) peak at $q = 1 \text{ \AA}^{-1}$ (Fig. 3l) verifies that the target film exhibits enhanced orientation, with stronger diffraction signals at azimuthal angles of 120° and 160°. This preferential growth direction likely enhances film performance by reducing defect state density, leading to a more ordered crystalline structure and improved film quality.

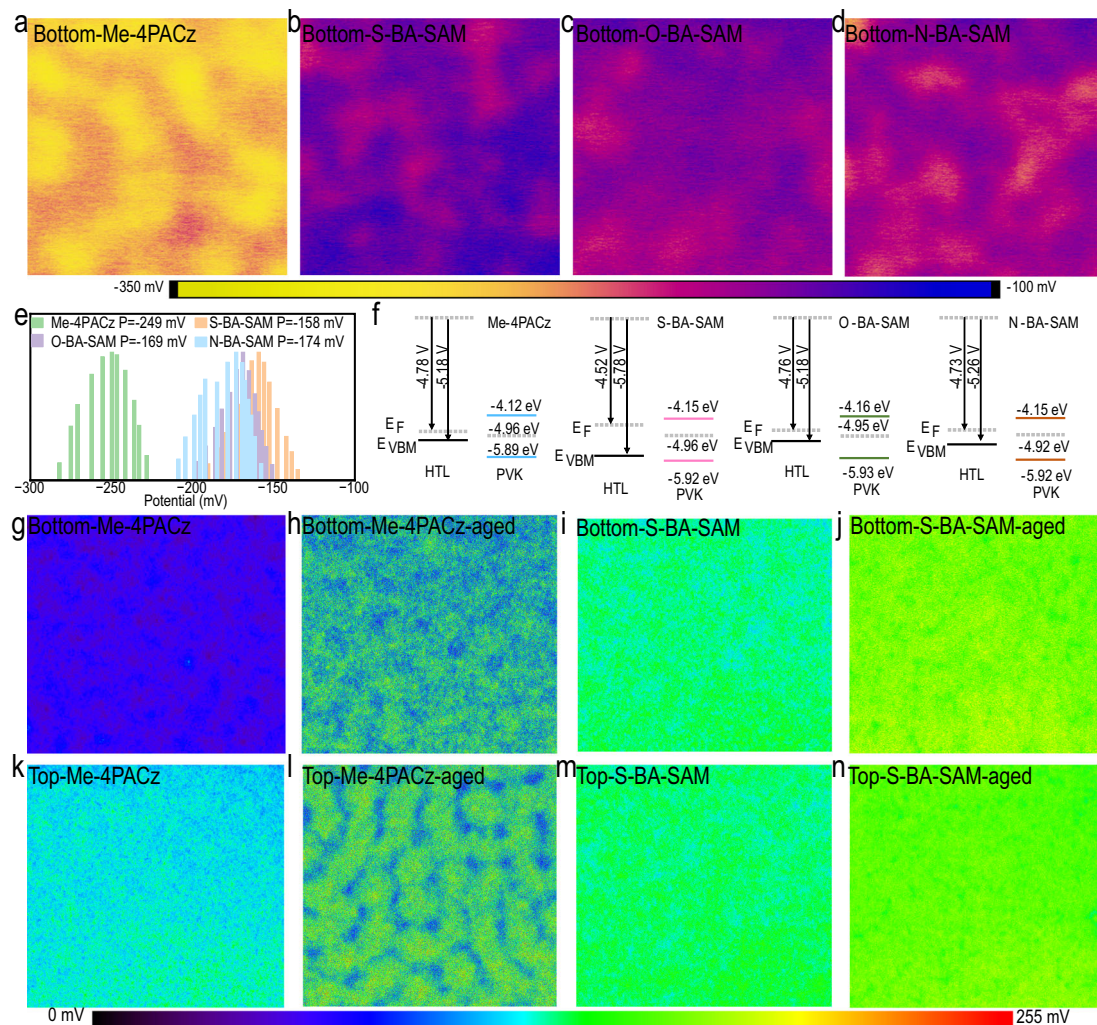


Fig. 4 | Effects of different SAMs on optoelectronic characteristics of perovskite films. KPFM images of the perovskite buried interface modified by (a) Me-4PACz, (b) S-BA-SAM, (c) O-BA-SAM, (d) N-BA-SAM. (e) CPD distribution histogram of the perovskite buried interface modified by Me-4PACz, S-BA-SAM, O-BA-SAM, N-BA-SAM. (f) Energy level diagrams of the SAM (Me-4PACz, S-BA-SAM, O-BA-SAM, N-BA-SAM)/perovskite. PL mapping of the bottom surface of perovskite films

on Me-4PACz (g) before and (h) after 5-h light aging; PL mapping of the bottom surface of perovskite films on S-BA-SAM (i) before and (j) after 5-h light aging; PL mapping of the top surface of perovskite films on Me-4PACz (k) before and (l) after 5-h light aging, PL mapping of the top surface of perovskite films on S-BA-SAM (m) before and (n) after 5-h light aging.

These are further confirmed by the lower trap densities ($1.73 \times 10^{16} \text{ cm}^{-3}$ vs. $1.63 \times 10^{16} \text{ cm}^{-3}$) and enhanced carrier transport mobilities ($7.92 \times 10^{-5} \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$ vs. $6.47 \times 10^{-5} \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$) extracted from the IV characteristics in Supplementary Fig. 39. Presented by the GIWAXS patterns and the azimuth profiles of O-BA-SAM, and N-BA-SAM modified perovskite film in Supplementary Fig. 40, it is seen that they both have fewer effects on perovskite morphology compared to S-BA-SAM.

As severe phase segregation in WBG perovskite would cause local lattice mismatch (between the I-rich and Br-rich regions) and the subsequent residual tensile stresses^{53,54}, the residual stresses within the perovskite films were then quantitatively analyzed using depth-resolved grazing incident X-ray diffraction (GIXRD) patterns. The stress (σ) was calculated based on the following equation $\sigma = \frac{-E}{2(1+\nu)} \frac{\pi}{180^\circ} \cot \theta \frac{\partial(2\theta)}{\partial \sin^2 \Psi}$ ⁵⁵, where E and ν respectively represent Young's modulus and Poisson's ratio of the perovskite film. From the 2θ - $\sin^2 \Psi$ plots (Supplementary Fig. 41) derived from the GIXRD patterns in Supplementary Fig. 42, the tensile stresses were determined to be 25, 22, 20, and 16 MPa for the control, O-BA-SAM, N-BA-SAM and S-BA-SAM treated WBG perovskite films, respectively. The released tensile stress could be accredited to the controllable crystallization of

Br/I halide, as previously discussed, eliminating the lattice mismatch issues. It is in turn illustrates higher intrinsic perovskite film stability.

Effects of mixed SAMs on optoelectronic characteristics of the WBG perovskite films

The surface potential of both buried and top perovskite films on different NiO_x/SAM substrates was further investigated by KPFM measurements, as illustrated in Fig. 4a–d and Supplementary Figs. 43–45. As summarized in Fig. 4e, the CPD of buried interfaces on mixed SAMs shifts upward by 91, 80, and 75 mV for S-BA-SAM, O-BA-SAM, and N-BA-SAM, respectively, as compared to that of the control film on Me-4PACz. The less negative CPD values are suggestive of more p-type characteristics of the buried interface properties, which is beneficial for hole conduction from perovskite to NiO_x/SAM HTL. Meanwhile, the narrower CPD distribution for the perovskite films treated with S-BA-SAM (full-width at half maximum FWHM of 25 mV), O-BA-SAM (FWHM of 26 mV), and N-BA-SAM (FWHM of 32 mV) compared to the control film (FWHM of 35 mV) implies improved film homogeneity. Similarly, in Supplementary Figs. 43–45, KPFM of the top surfaces reveals comparable CPD peaks of 114, 96, 87, and 149 mV for the Me-4PACz, S-BA-SAM, O-BA-SAM, and N-BA-SAM based perovskite films, respectively.

These values are consistent with the Fermi level of perovskite extracted from UPS measurements (shown in Fig. 4f). The $\text{NiO}_x/\text{S-BA-SAM}$ system demonstrates a well-matched highest occupied molecular orbital (HOMO) level with the perovskite layer (Fig. 4f), minimizing interfacial energy losses. Notably, S-BA-SAM-based perovskite film shows much narrower FWHM than those of others, further indicating enhanced film uniformity. Under continuous light illumination aging for 6 h, the control film exhibits a broadening of CPD distributions and a significant shift to less positive values (from 114 mV to 75 mV), which might signify the formation of higher work function PbI_2 regions (5.9 eV). This reflects the degradation of control WBG perovskite into I-rich and Br-rich regions. In contrast, the film on $\text{NiO}_x/\text{S-BA-SAM}$ substrate shows only minor changes in surface potential characteristics after light aging (Supplementary Fig. 44c), evidencing suppressed photo-induced phase segregation.

Steady-state and time-resolved photoluminescence (PL and TRPL) measurements were conducted on the four WBG perovskite films. As shown in the PL spectra (Supplementary Fig. 46), the PL quenching increases with mixed SAM substrates, following the trending of incremented electron capture effects of the fused ring (i.e., benzothiophene > pyridine > benzofuran), which suggests a reduction in trap-assisted nonradiative recombination losses. Consistently, bi-exponentially fitting of the TRPL spectra (Supplementary Fig. 46b, c) gives remarkably elongated carrier recombination lifetime from 26.1 (control) to 483.9, 247.2, and 238.72 ns, respectively, for S-BA-SAM, O-BA-SAM, and N-BA-SAM, in line with the PL spectra. Fitting details are shown in Supplementary Table 7. The reduced defects and recombination could be attributed to the homogeneous crystallization of Br/I regions, suppressing the formation of narrow bandgap recombination sites.

Light stability of the WBG perovskite on different NiO_x/SAM substrates was subsequently assessed. As shown in the evolution of the PL spectra under continuous light illumination (Supplementary Fig. 47), the corresponding 2D pseudo-color images are plotted in Supplementary Fig. 48. It is evident that the PL spectrum of the control film exhibits significant redshifts during the first 90 min of illumination, indicating the onset of phase segregation. This suggests that photo-generated charge carriers tend to migrate to low-energy I-rich regions followed by being quenched. Upon extended illumination for 5 h, additional shoulders appear around 760–810 nm, manifesting severe phase segregation of the I/Br mixed halides into distinct Br-rich and I-rich regions. Contrastingly, the WBG perovskite on $\text{NiO}_x/\text{S-BA-SAM}$ displays only a marginal redshift without the appearance of lower-energy PL shoulders, evidencing the significantly suppressed phase segregation and enhanced light stability of the target film⁵⁶.

Photoluminescent properties of the perovskite films at the microscopic scale were examined using confocal PL mapping, before and after light illumination. Both top and buried surfaces were analyzed. Comparing PL mapping images of the fresh perovskite films (bottom interface) in Fig. 4g–j, the target film on $\text{NiO}_x/\text{S-BA-SAM}$ exhibits an overall enhancement in PL intensity across a $5\ \mu\text{m} \times 5\ \mu\text{m}$ area, whereas the control film exhibits obvious inhomogeneity with “wrinkle” morphologies (Supplementary Fig. 49). The ameliorated buried interfacial photoluminescent characteristics could be attributed to the strong anchoring of S-BA-SAM on NiO_x and a complete surface coverage, which promotes homogeneous crystallization of Br/I halides. After 5 h of continuous light illumination, the buried interface undergoes severe degradation, with coarsened film morphology (Supplementary Fig. 49) and reduced PL intensities (Fig. 4h). Contrastingly, the target buried film retains its intact film morphology and uniform PL intensity distribution, as illustrated in Fig. 4i, j and Supplementary Fig. 49. Similarly, the top surface of perovskite film on $\text{NiO}_x/\text{S-BA-SAM}$ substrate also demonstrates preferable photoluminescent properties and light stability over those of the control film, as shown in Fig. 4k–n and Supplementary Fig. 50.

The PV performance of WBG single-junction PSCs

Motivated by the balanced Br/I halide crystallization, enhanced HTL/perovskite interfacial stability, and improved optoelectronic properties of perovskite films achieved through the mixed SAM strategy, single-junction WBG PSCs were fabricated using the device architecture of NiO_x/SAM or mixed SAM/ WBG/ $\text{C}_{60}/\text{ALD SnO}_2/\text{Ag}$, as shown by the inset of Fig. 5a. J-V characteristics of these cells were recorded under simulated 1 sun illumination at an intensity of $100\ \text{mW}/\text{cm}^2$ (AM 1.5 spectrum). The optimal molar ratio of BA-SAMs mixed with Me-4PACz was determined to be 4:1 (BA-SAM: Me-4PACz), as illustrated in Supplementary Fig. S1. Figure 5a compares the J-V characteristics of the champion devices of control and mixed SAM-based WBG PSCs. From the photovoltaic parameters summarized in Supplementary Table 8, S-BA-SAM based device shows remarkably enhanced PCE from 18.9% to 20.1%, along with improvements in V_{OC} from 1.28 V to 1.30 V, J_{SC} from 18.1 to 18.2 mA cm^{-2} and FF from 82.08% to 84.8%, compared to those of the control device. As seen in Supplementary Table 9, the N-BA-SAM and O-BA-SAM modified devices show progressively increased PCE over that of the control device, in line with the electron capture effect of the boron acid-SAMs. Noted that PSCs based on pure S-BA show relatively lower PCE of 19.0% than that of mixed S-BA-SAM based ones, with V_{OC} of 1.28 V, J_{SC} of 18.1 mA cm^{-2} and FF of 82.1% (Supplementary Fig. S2 and Supplementary Table 10). This could be explained by the previous quantum chemical calculations that a mixture of S-BA and Me-4PACz affords a higher degree of charge delocalization in frontier orbitals than pure S-BA, facilitating hole transport. The assumptions are also testified by the conductivity (Supplementary Fig. 29) and mobility measurements (Supplementary Fig. 39). The steady-state PCE output (SPO) tracking results (Fig. 5b and Supplementary Fig. S3) show steady-state PCEs of 17.9%, and 19.3%, 18.7%, and 18.2% for the control, S-BA-SAM, O-BA-SAM, N-BA-SAM based devices, respectively, corroborating the reliability of the J-V characteristics. Integration of the external quantum efficiency (EQE) spectra of the PSCs (Fig. 5c) yields J_{SC} values of 17.3 mA cm^{-2} and 17.7 mA cm^{-2} for the control and target devices, respectively, aligning well with the J-V results. Figure 5d, e demonstrates the statistical performance of 20 individual cells for both PSCs, detailed parameters were in Supplementary Tables 11 and 12, which shows preferable reproducibility of the photovoltaic parameters (V_{OC} , PCE) of the target devices over those of control ones, likely due to improved homogeneity and reduced defects in the polycrystalline film.

In order to gain deeper insight into the significant V_{OC} and FF enhancement with mixed S-BA-SAM strategy, charge carrier transport and recombination dynamics in the devices were systematically studied. The nonradiative recombination at interfacial contacts and corresponding energy losses were firstly assessed by photoluminescence quantum yield (PLQY) of perovskite and HTL/perovskite films. It is seen from Fig. 5f and Supplementary Fig. S4 that PLQY of the control perovskite film increases from 0.238% to 0.255%, 0.248%, and 0.243% with S-BA-SAM, O-BA-SAM, N-BA-SAM modification, consistent with the PL results. While, in contact with $\text{NiO}_x/\text{Me-4PACz}$ HTL, the PLQY value of the control film substantially reduces to 0.150%, suggesting the remarkable nonradiative recombination and energy losses at the HTL/perovskite interface. This could be ascribed to the numerous voids and defects formed at the buried interface as proved in previous context. By refining the buried contacts, PLQY incremented to 0.247%, 0.223%, and 0.200% for the target $\text{NiO}_x/\text{S-BA-SAM}$, O-BA-SAM, and N-BA-SAM/perovskite film. Furthermore, in Fig. 5f, we estimate the quasi-Fermi level splitting (QFLS) and implied open circuit voltage (iV_{OC}) ($\text{QFLS} = e \times iV_{\text{OC}}$), e is the elementary charge^{24,57}. The QFLS of the Me-4PACz/PVK sample decreases significantly to 1.296 eV compared to 1.322 eV of pure perovskite, while the QFLS of S-BA-SAM/WBG sample reduces slightly from 1.326 to 1.322 eV. It could be attributed to the improved energy level arrangement and passivated surface defects of

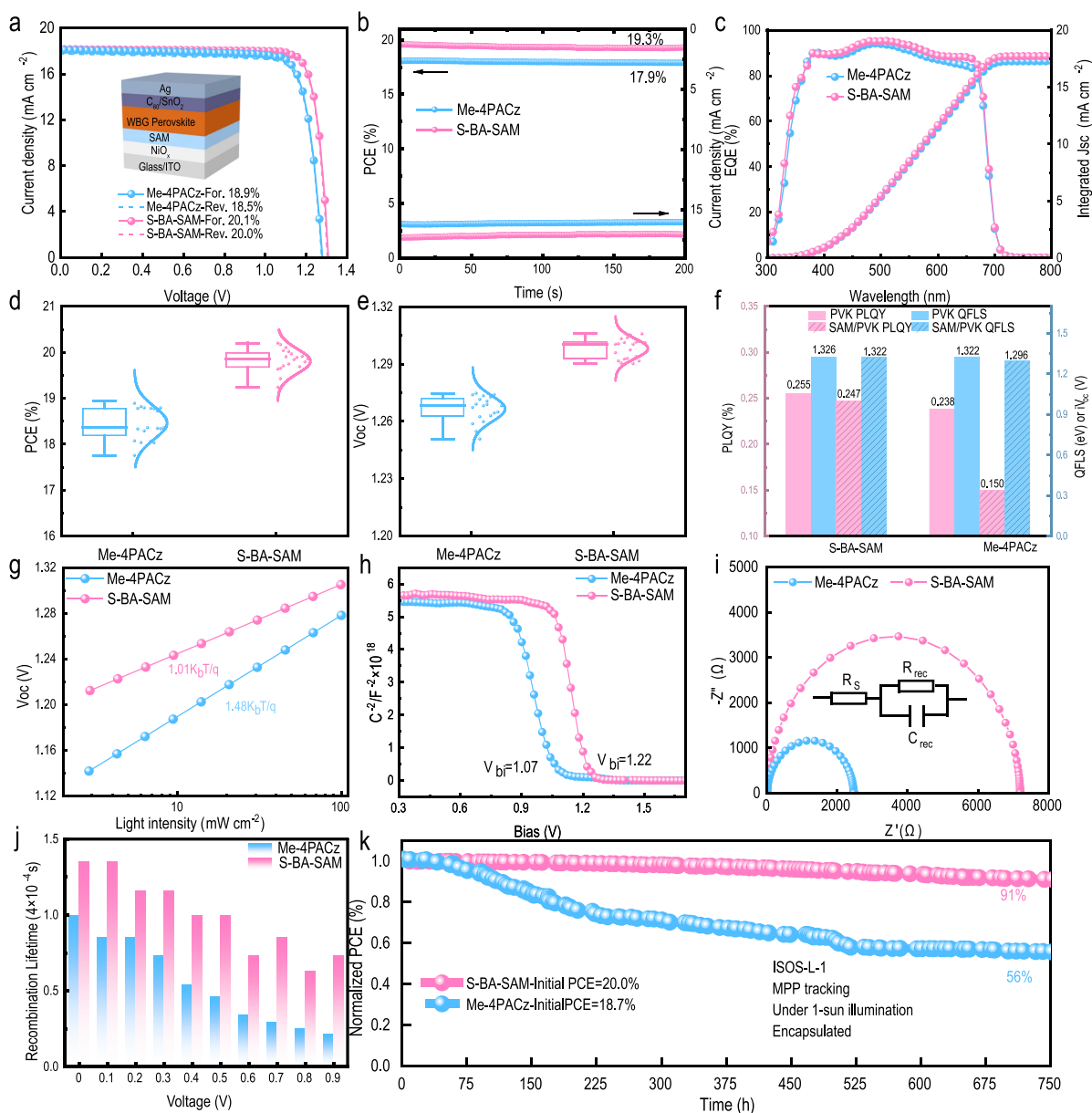


Fig. 5 | WBG photovoltaic performance and stability. (a) J-V characteristics of champion devices based on Me-4PACz and S-BA-SAM. (b) SPO of champion devices based on Me-4PACz and S-BA-SAM. (c) EQE spectra of champion devices based on Me-4PACz and S-BA-SAM. Statistical (d) PCE and (e) V_{OC} of the device based on Me-4PACz and S-BA-SAM. (f) PLQY and QFLS or iV_{OC} values of perovskite film based on Me-4PACz and S-BA-SAM. (g) V_{OC} versus light intensity of perovskite

film based on Me-4PACz and S-BA-SAM. (h) Mott-Schottky plots of devices based on Me-4PACz and S-BA-SAM. (i) Nyquist plots of devices based on Me-4PACz and S-BA-SAM. (j) Recombination lifetime of the devices extracted from the middle-frequency region of the Nyquist plots of devices based on Me-4PACz and S-BA-SAM. (k) MPP tracking of encapsulated control and target devices under 1 sun illumination.

perovskite with S-BA-SAM treatment, which minimizes non-radiative recombination losses at the interface.

The carrier recombination process under light illuminations was further evaluated by light intensity (P_{light}) dependent V_{OC} . From Fig. 5g and Supplementary Fig. 55, slopes of V_{OC} dependence on light intensity (P_{light}) of the control, O-BA-SAM modified devices, N-BA-SAM modified devices, and target devices are determined to be 1.01, 1.44, 1.45, and 1.48 k_BT/q, respectively, indicative of a suppressed trap-assisted recombination in the target device. The Mott-Schottky characteristics of the PSCs were also studied. From Fig. 5h, the mixed SAM treatment notably increments the built-in potential (V_{bi}) of the device from 1.07 V for the control device to 1.22 V for the target device. It arises from the reduced trap-assisted recombination and suppressed

interfacial energy losses, conforming to the improved V_{OC} of S-BA-SAM-based PSCs. Additionally, based on the equation $\frac{dC^{-2}}{dV} = \frac{2}{A^2 q \epsilon_0 N_t}$ ⁵⁸, charge carrier density (N_t) decreases from $3.8 \times 10^{14} \text{ cm}^{-3}$ to $2.9 \times 10^{14} \text{ cm}^{-3}$ with S-BA-SAM modification, manifesting accelerated charge transport thus lower charge accumulation at the HTL/perovskite interface. Electrochemical impedance spectroscopy (EIS) of the devices under external voltages of 0.0–0.9 V were further measured⁵⁹. EIS spectra (0.9 V) of the control and target PSCs are compared in Fig. 5i, which shows higher recombination resistance of the target device (7158 Ω) over that of the control one (2463 Ω). Based on the EIS spectra under different external biases (Supplementary Fig. 56), the carrier recombination lifetimes of the perovskite films are

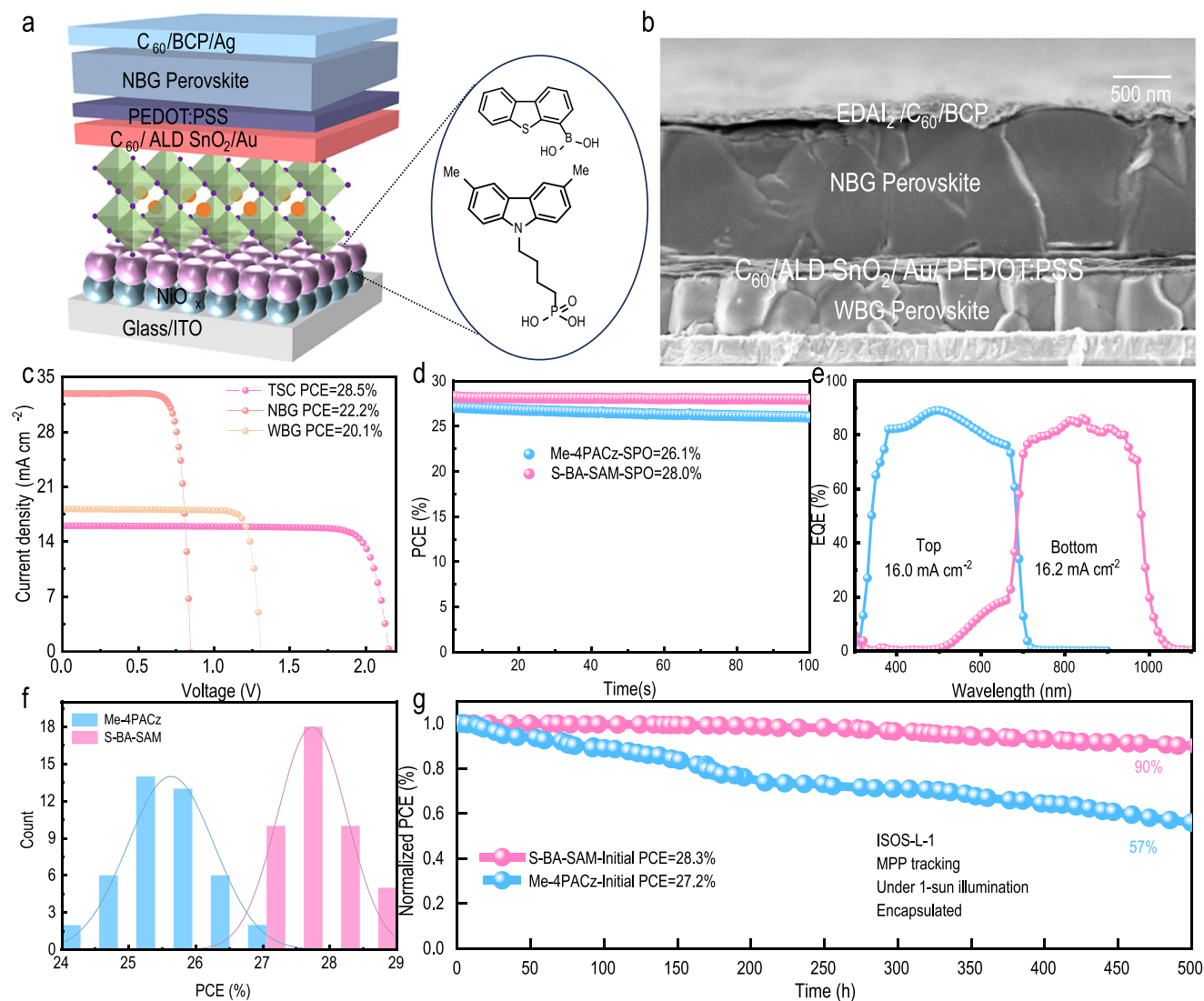


Fig. 6 | Tandem solar cells photovoltaic performance analysis. (a) The device architecture of 2-T tandem solar cells. (b) Cross-sectional SEM images of the devices. (c) J–V characteristics of three different devices: single-junction NBG PSC, single-junction WBG PSC, and tandem solar cells. (d) SPO of the champion 2-T TSCs

based on Me-4PACz and S-BA-SAM under working conditions with $100\ mW\ cm^{-2}$ irradiation. (e) EQE spectra of TSCs. (f) Statistic histogram of PCE. (g) MPP tracking of encapsulated control and target devices under 1 sun illumination.

extracted and compared in Fig. 5j and Supplementary Fig. S7. The mixed SAM treatment delivers an overall prolonged carrier recombination lifetimes than those of the control ones. These results further evince the reduced probability for non-radiative recombination in the target device.

Effects of the buried layer modification on the operational stability of the WBG PSCs were investigated by tracking the maximum power point (MPP) under continuous 1 sun illumination. All the devices for testing were encapsulated. As demonstrated in Fig. 5k, the control device experiences rapid degradation to only 56% of the initial PCE after 750 h illumination. This is likely due to the combined effects of the weak interfacial contacts at NiO_x /SAM and the phase segregation within the WBG perovskite film. Contrastingly, the target WBG device exhibits drastically enhanced resilience to light illumination, with 91% of the initial PCE retained after 750 h aging (under ISOS-L-1 protocol). The thermal effect has also been taken into consideration by conducting tests under ISOS-L-2 protocol (1sun MPP tracking under $85^\circ C$). As illustrated by Supplementary Fig. S8a, the WBG subcell retains 86% of the initial PCE after 200 h MPP tracking, while that of the control device drops to 63% of the initial PCE. The stronger

resilience under both light and thermal stresses could be accredited to improved homogeneity of co-SAM distribution and their robust anchoring on NiO_x , ameliorating the interfacial stability. Suppressed phase segregation of WBG film also contributes to the enhanced stability.

Performance and stability of all-perovskite TSCs

The configuration of the TSCs is ITO/ NiO_x /SAM or Mixed SAM/WBG perovskite/ C_{60} /SnO₂/Au/PEDOT:PSS/NBG perovskite/ C_{60} /BCP/Ag (Fig. 6a). Cross-sectional SEM image of the tandem device is presented in Fig. 6b, which shows the thickness of the WBG and NBG perovskite films to be ~1100 and ~400 nm, respectively. The monolithic 2T all-perovskite TSCs were further fabricated by integrating the WBG perovskite as the top subcell (described above) with 1.25 eV NBG perovskite as the bottom subcell (see Methods for details). As illustrated in Fig. 6c, the NBG PSCs show PCE of 22.2%, with V_{OC} of 0.85 V, J_{SC} of $32.9\ mA\ cm^{-2}$ and FF of 79.8%. With S-BA-SAM modification on NiO_x , the champion device achieves an ameliorative PCE of 28.5%, with V_{OC} of 2.15 V, J_{SC} of $16.0\ mA\ cm^{-2}$ and FF of 83.0% (Fig. 6c), and PCE was stabilized at 28.0% (Fig. 6d). While the control device exhibits a much

inferior PCE of 26.1%. Integrated from EQE spectra, the bottom NBG and top WBG subcells show well-matched integrated J_{SC} of 16.0 and 16.2 mA cm⁻² (Fig. 6e), respectively, in good agreement with the J_{SC} values obtained from J-V characteristics. A total of 43 all-perovskite TSCs were fabricated (in Fig. 6f), demonstrating the preferable reproducibility of the target device. Furthermore, accredited to the significantly improved light and thermal stability of WBG subcells, the target TSC devices also demonstrate notably elongated operational stability than that of the control device, with only 10% degradation in PCE after 500 h tracking under ISOS-L-1 protocol and 20% decrement in PCE after 150 h aging under ISOS-L-2 protocol (Fig. 6g and Supplementary Fig. S8b).

Discussion

In summary, we have demonstrated that buried interface engineering by using a mixed SAM (S-BA-SAM: Me-4PACz = 4:1, molar ratio) has a substantial impact on the performance of WBG subcells and TSCs. By exploring the interaction between the BA anchoring group of BA-SAM and NiO_x, as well as the interaction between conjugation π cores of BA-SAM and FA⁺ cation, we successfully improve the interfacial contact between HTL/perovskite and suppress the notorious phase segregation in WBG perovskite film. As a result, significant improvements in key parameters such as V_{OC} and FF were achieved, delivering a meliorative PCE of 20.1% for the WBG subcell and 28.5% for all-perovskite TSCs. Both modified devices demonstrate significantly improved light and thermal stabilities compared to the control ones. Overall, our research into buried interface engineering provides valuable insights for further advancements and optimization in the field of perovskite photovoltaics.

Methods

Materials

Formamidine iodide (FAI, 99.99%), methylammonium chloride (MACl 99.99%), lead (II) iodide (PbI₂ 99.9%), cesium iodide (CsI), lead bromide (PbBr₂), lead (II) chloride (PbCl₂ 99.9%), nickel oxide (NiO_x) and patterned ITO substrates were purchased from Advanced Election Technology CO., Ltd.. Isopropanol (IPA, 99.9%), chlorobenzene (CB, 99.9%), N, N-dimethyl formamide (DMF, 99.8%), diethyl ether (DE), and dimethyl sulfoxide (DMSO, 99.7%) were obtained from Beijing J&K Scientific Ltd.. Tin (II) fluorine (SnF₂, 99.9%) and ammonium thiocyanate (NH₄SCN, 99.9%) and lead (II) thiocyanate (Pb(SCN)₂, 99.9%) and 4-Dibenzothienyl boronic acid (S-BA) were purchased from Sigma-Aldrich. PEDOT: PSS (CLEVIOS P VP Al 4083) was purchased from Heraeus. Methylammonium iodide (MAI), cesium iodide (CsI), lead bromide (PbBr₂), and ethanediamine dihydroiodide (EDAI)₂ were supplied from Xi'an Polymer Light Technology Corporation. 4-(Dibenzofuranyl) boronic acid (O-BA) was purchased from Macklin and 9H-Carbazol-1-yl) boronic acid (N-BA) was purchased from Bide pharm.

WBG FA_{0.8}Cs_{0.2}PbI_{1.8}Br_{1.2} perovskite solution preparation

Dissolve 221.3 mg PbI₂, 264.2 mg PbBr₂, 165.1 mg FAI, 6.67 mg PbCl₂, 62.4 mg CsI, and 1.62 mg MACl in 1 mL DMF and DMSO (DMF: DMSO = 4:1, v/v) mixed solution to prepare of (1.2 M) perovskite precursors. Finally, the solution was prepared by filtering the solution with 0.22 μ m polytetrafluoroethylene (PTFE) membrane.

NBG FA_{0.6}MA_{0.3}Cs_{0.1}Pb_{0.5}Sn_{0.5}I₃ perovskite solution preparation

Dissolve 507.10 mg PbI₂, 409.77 mg SnI₂, 224.56 mg FAI, 104.92 mg MAI, 57.16 mg CsI, 17.23 mg SnF₂ and 3.14 mg NH₄SCN in 1 mL DMF and DMSO (DMF: DMSO = 3:1, v/v) mixed solution to prepare (2.2 M) perovskite precursors. Finally, the solution was prepared by filtering the solution with 0.22 μ m polytetrafluoroethylene (PTFE) membrane.

Device fabrication

WBG perovskite solar cell fabrication. First, the patterned ITO substrates were ultrasonic cleaned with detergent, deionized water, acetone, and isopropyl alcohol successively for 30 min, then dried with N₂ and treated with plasma for 5 min. A layer of NiO_x nanocrystals (10 mg ml⁻¹ in H₂O) was first coated on an ITO substrate spinning at 1500 rpm for 30 s and then annealed at 150 °C in air for 10 min. After cooling, the substrate is immediately transferred to a N₂-filled glove box. Next, a self-assembled monolayer Me-4PACz or mixed SAM (0.5 mg ml⁻¹ ethanol) was spinning on NiO_x substrate at 3000 rpm for 30 s. The WBG perovskite film is deposited using a two-step spin-coating process: 500 rpm for 2 s and 4000 rpm for 60 s, and in a second spin-coating step, DE is dripped onto the spinning substrate at 25 s. The substrate is then transferred to a hot plate and heated at 100 °C for 10 min. After cooling to room temperature, the substrate is transferred to the evaporation system and a 25 nm C₆₀ film is subsequently deposited. A layer of ALD SnO₂ with a thickness of 20 nm was deposited on the C₆₀ film, and then 100 nm Ag was deposited by thermal evaporation.

All-perovskite tandem solar cell fabrication. In the all-perovskite tandem solar cell, NiO_x solution, SAM solution and perovskite solution were spin-coated in sequence, and then 25 nm of C₆₀ was deposited by thermal evaporation. A layer of ALD SnO₂ with a thickness of 20 nm was deposited on the C₆₀ film, and then 1 nm Au was deposited by thermal evaporation. Then took out the glovebox and spin-coated the PEDOT layer at 4000 rpm for 30 s and annealed in air at 120 °C for 20 min. The film is then transferred to an N₂ atmosphere glove box for further spin coating. The prepared NBG precursor solution was spin-coated in a two-step process: 1000 rpm for 10 s, 4000 rpm for 40 s, drops of anti-solvent (CB) in 30 s, and then annealed at 100 °C for 10 min in the glove box. Then, the EDAI₂ (IPA with a concentration of 1.0 mg ml⁻¹) layer was spin-coated at 4000 rpm for 20 s, then annealed at 100 °C for 1 min, and then sequentially deposited 25 nm C₆₀, 6 nm BCP, and 100 nm Ag by thermal evaporation.

Characterization and measurements

Computational details. All the calculations are performed in the framework of the density functional theory with the projector-augmented plane-wave method, as implemented in the Vienna ab initio simulation package⁶⁰. The generalized gradient approximation proposed by Perdew, Burke, and Ernzerhof is selected for the exchange-correlation potential⁶¹. The long-range van der Waals interaction is described by the DFT-D3 approach⁶². The plane wave cut-off energy of 400 eV is adopted, the energy convergence accuracy is set to 1 $\times 10^{-5}$ eV/atom, and the force acting on each atom is not greater than 0.1 eV/Å. The Brillouin zone is integrated using a 2 $\times$ 1 $\times$ 1 k-point grid. All quantum chemical calculations are performed using Gaussian09. For the initial structure, use the B3LYP density functional with the 6-311 G basis set. The NiO_x model was optimized geometrically through density functional theory (DFT) calculations performed with the CP2K⁶³. package, utilizing a mixed Gaussian and plane wave (GPW) basis set. The calculations incorporated the Perdew-Burke-Ernzerhof (PBE) exchange-correlation functional, DFT-D3 dispersion corrections, and dipole corrections⁶⁴ necessary for periodic boundary conditions along the direction perpendicular to the surface. The valence electron wave functions were expanded in a double- ζ Gaussian basis set with polarization functions (DZVP)⁶⁵. An energy cutoff of 400 Ry was applied for the electron density expansion in the GPW method.

Molecular dynamics (MD) simulations were conducted using the GROMACS (version 2021.6) simulation package with the General Amber Force Field (GAFF2)⁶⁶. RESP charges were calculated using the Multiwfn program⁶⁷. The molecules were positioned atop the NiO_x surface, modeled using a universal force field encompassing the entire periodic table. Following thousands of steps of energy minimization, a

10 ns equilibration was performed at 300 K with position restraints applied to the NiO_x surface. The production runs were extended for an additional 10 ns under the NVT ensemble, with snapshots recorded every 1 ps. Temperature control at 300 K was achieved using the Nose-Hoover thermostat. A non-bonded interaction cutoff of 1.0 nm was implemented, and the Particle Mesh Ewald (PME) method with a Fourier spacing of 0.1 nm was applied to handle long-range electrostatic interactions⁶⁸.

Device characterization. The Newport Oriel sol3A 450 W solar simulator was used to test the current density versus voltage (J–V) curves and stabilized power output (SPO) under AM 1.5 G. For J–V scanning of all cells, we place aperture masking masks in front of the solar cells to ensure an effective area of 0.06 cm². A J–V scan was performed on the WBG perovskite solar cell with a scan rate of 0.1 V s^{−1} and a delay time of 50 ms. The forward scanning range is −0.2–1.4 V, and the reverse scanning range is 1.4–−0.2 V. A J–V scan was performed on the Sn–Pb perovskite solar cell with a scan rate of 0.1 V s^{−1} and a delay time of 50 ms. The forward scanning range is −0.1–1.0 V, and the reverse scanning range is 1.0–−0.1 V. A J–V scan was performed on the all-perovskite tandem solar cell with a scan rate of 0.1 V s^{−1} and a delay time of 50 ms. The forward scan is −0.2–2.2 V, and the reverse scan is 2.2–−0.2 V. The area of the solar cell under test is 0.0116 cm². The solar cell quantum efficiency test system (Elli Technology Taiwan) was used to measure the EQE spectra of devices. The Mott Schottky curves and the impedance spectroscopy (IS) were determined with the Chenhua CHI760E electrochemical workstation. EQE measurements were measured by applying external voltage/current sources through the PSCs with a REPS measurement instrument (Enlitech). Operational stability tests of WBG/tandem solar cells were performed at maximum power point (MPP) in N₂ environment under AM1.5 xenon lamp illumination (100 mW cm^{−2}, without UV filter).

Film characterization. The polarizer is made of ZnSe, which limits the low-end spectral range to around 650 cm^{−1}. In these experiments, a different background is required for each polarization position used. For example, if you are going to collect spectra at 0° and 90°, corresponding background spectra are required at 0° and 90°. The spectral resolution was set to 4 cm^{−1}, the aperture was set to 4 mm and spectra were acquired by averaging 256 scans. The X-ray photoelectron spectroscopy (XPS) was performed by a multifunctional photoelectron spectrometer (Axis Ultra DLD, Kratos) under ultrahigh vacuum (3.0 × 10^{−8} Torr) with a non-monochromatic He-I excitation (21.22 eV). The in-situ dynamic absorption spectrum is measured by the multi-spectral analysis equipment proposed by spectral microvision. Through the combination of the spinning instrument, LED lamp, spectrometer (ATP2002), sample table, and the display screen of the spinning instrument, the test material is evenly coated on the glass substrate on the sample platform, the instrument parameters are set, and the LED light source is irradiated vertically on the sample through the optical fiber. The transmission spectrum was detected by the spectrometer, the spectral resolution reached 0.01 nm, and the time resolution reached 1 ms. The nucleation and crystallization process of perovskites were analyzed by spectral data, and the morphological characteristics of the films were detected and revealed in real-time. The Time-of-Flight Secondary Ion Mass Spectrometry (TOF-SIMS) analysis was performed using a dual-beam approach. Primary ion bombardment was carried out using Bismuth (Bi³⁺) ions at an energy of 30 keV and a current of 45 degrees per nanoampere. Secondary ion detection was facilitated by Cesium (Cs⁺) ions at an energy of 1 keV and a current of 80 nanoamps, with the secondary ion beam aligned at 45 degrees to the primary ion path. Additionally, a flood gun was employed to neutralize the charge on the sample surface, ensuring accurate mass resolution and ion yield. PLQY measurements were characterized by a system with an integrating sphere and an excitation

wavelength of 365 nm. The fixed light intensity of 100 mW cm^{−2} was used for the PLQY measurements. The perovskite bottom interface was characterized with XPS, SEM, KPFM, QFSL, PL Mapping, and the thin film preparation procedure was shown in Supplementary Fig. S9.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The data generated in this study are provided in the Supplementary Information/Supplementary Data/Source Data file. Source data are provided with this paper.

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Author contributions

C.L. conceived the idea and guided the work. J.W., R.T., Y.M., and Y.B. designed the experiments, fabricated the perovskite films and devices, and analyzed measured results. B.J. calculated adsorption energy, electron localization function, differential charge density, quantum chemical, and molecular dynamics. Y.M., C.X., X.L., B.H., Y.W., M.Y., H.P., Z.S., and S.Z. helped the characterizations; K.S. calculated the adsorption energy; Z.G., C.L., and R.T. helped to revise the manuscript. All authors discussed the results and commented on the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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